

Postfix expression evaluation :

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1) Start scanning the postfix expression from left to right

Postfix exp : 59+4862/-*-173-5+

2) If current element is an operand

Current element : +

Push it onto the stack

Else

Op1 : -6

If current element is an operator

Op2 : 1

- Pop two elements (operands) from the stack.

Result :

- 1st popped element will be 2nd operand : op2

- 2nd popped element will be 1st operand : op1

- Perform cur element operation between op1 and op2

Result = -5

Result = op1 opr op2

- Push back result onto the stack

3) Repeat the above steps till end of postfix exp

4) Pop final result from the stack.

Stack

Prefix exp evaluation

Start scanning the prefix expression from right to left

If operand, push it on the stack.

If operator, pop two elements from the stack

1st popped element will be first operand.

2nd popped element will be second operand.

Perform the operation and push the result on the stack.